Simulating Age-Period-Cohort Data

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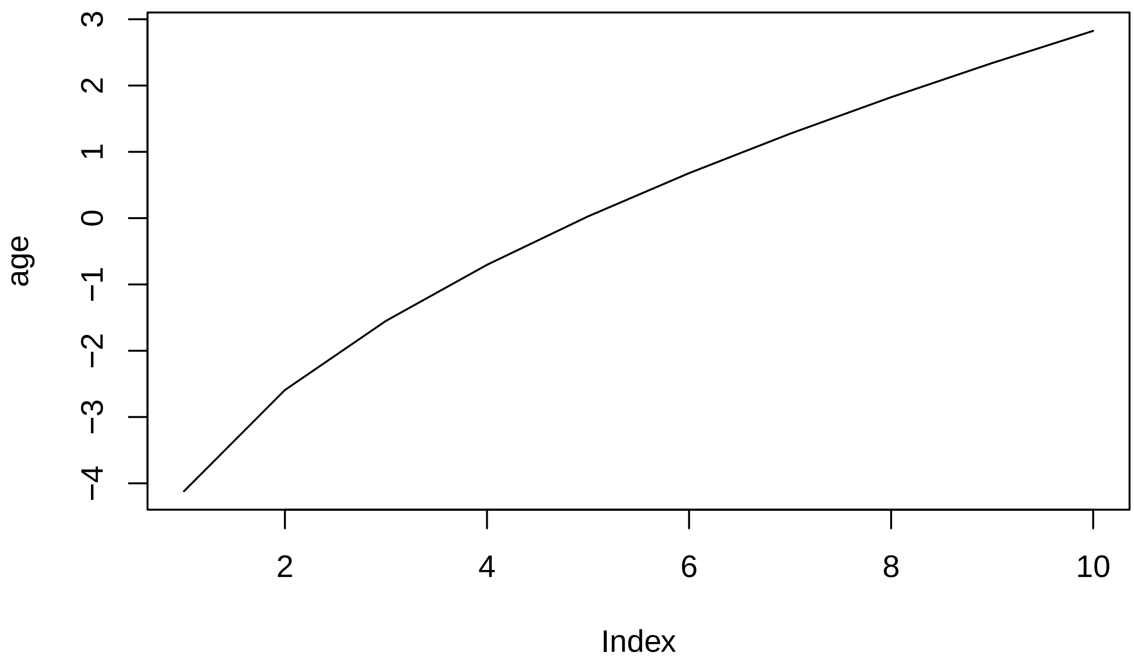
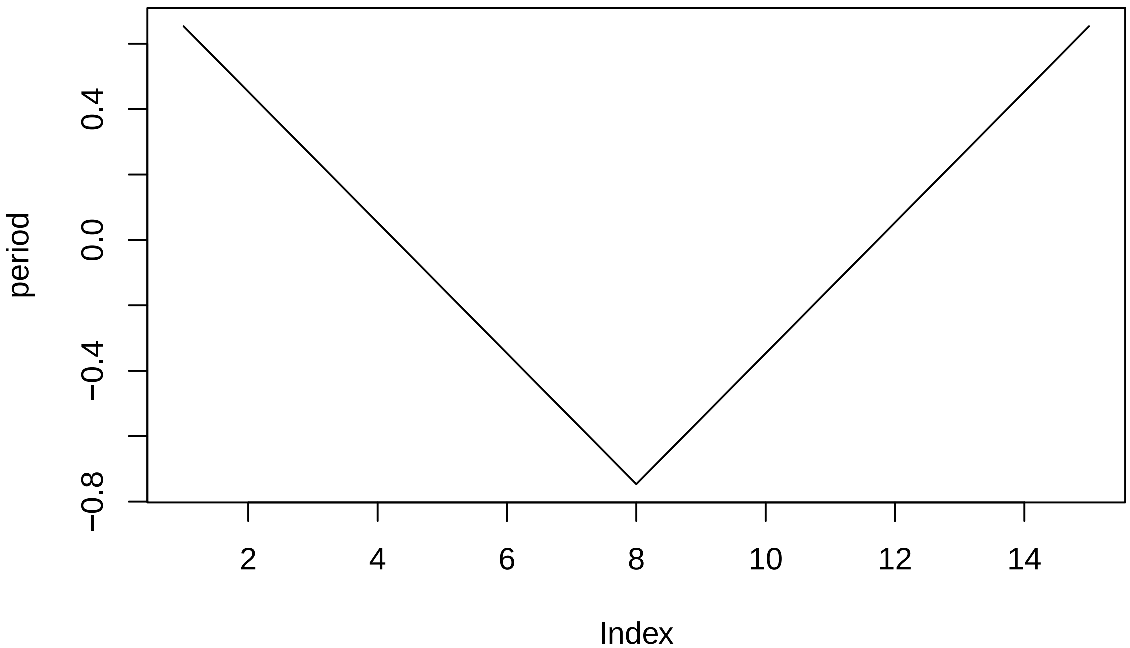
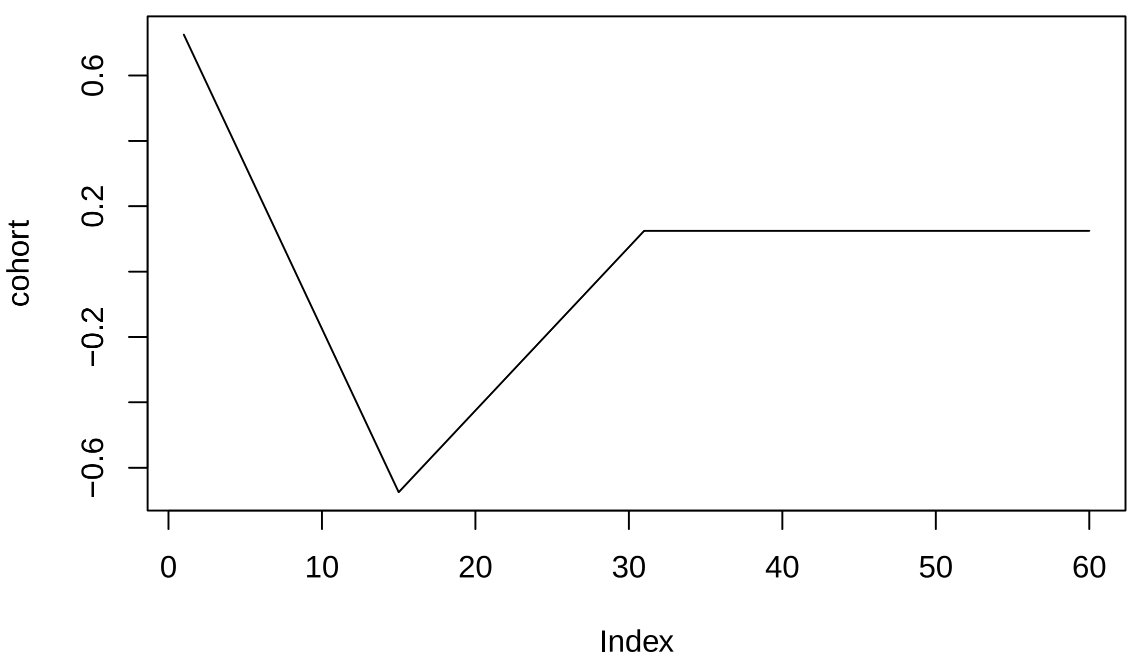
```
age=2*sqrt(seq(1,20,length=10))
age<- age-mean(age)
plot(age, type="l")
```




```
period=15:1
period[8:15]<-8:15
period<-period/5
period<-period-mean(period)
plot(period, type="l")
```




```r
periods_per_agegroup=5
number_of_cohorts <- periods_per_agegroup*(10-1)+15
cohort<-rep(0,60)
cohort[1:15]<-(14:0)
cohort[16:30]<- (1:15)/2
cohort[31:60]<- 8
cohort<-cohort/10
cohort<-cohort-mean(cohort)
plot(cohort, type="l")
```



```r
simdata<-apcSimulate(-10, age, period, cohort, periods_per_agegroup, 1e6)
print(simdata$cases)
```

```
##       [,1] [,2] [,3] [,4] [,5] [,6] [,7] [,8] [,9]
##  [1,]    1    7   21   49   67  125  160  366 1133
##  [2,]    2    7   19   49   76   94  155  311  810
##  [3,]    2    6   22   43   56   78  130  224  640
##  [4,]    1    4   13   32   46   82  107  183  452
##  [5,]    1    0    9   14   42   72   95  127  346
##  [6,]    0    2    3   10   21   56   87  113  233
##  [7,]    0    0   11   11   20   42   70   76  206
##  [8,]    0    2    3   11   37   41   59   84  123
##  [9,]    0    0    3   10   39   48   70   84  161
## [10,]    0    4    1   19   32   56  100  134  176
## [11,]    1    5   10   11   43   99  127  157  220
## [12,]    0    1   10   27   55   92  135  203  311
## [13,]    2    2   13   37   67  129  202  265  374
## [14,]    2    7   17   44   82  149  262  390  484
## [15,]    1    4   25   48  112  176  316  473  609
##       [,10]
##  [1,]  3084
##  [2,]  2229
##  [3,]  1663
##  [4,]  1242
##  [5,]   887
##  [6,]   671
##  [7,]   494
##  [8,]   361
##  [9,]   398
## [10,]   446
## [11,]   490
## [12,]   541
## [13,]   614
## [14,]   708
## [15,]   743
```

```r
simmod <- bamp(cases = simdata$cases, population = simdata$population, age = "rw1",
period = "rw1", cohort = "rw1", periods_per_agegroup =periods_per_agegroup)
```

```r
print(simmod)
```

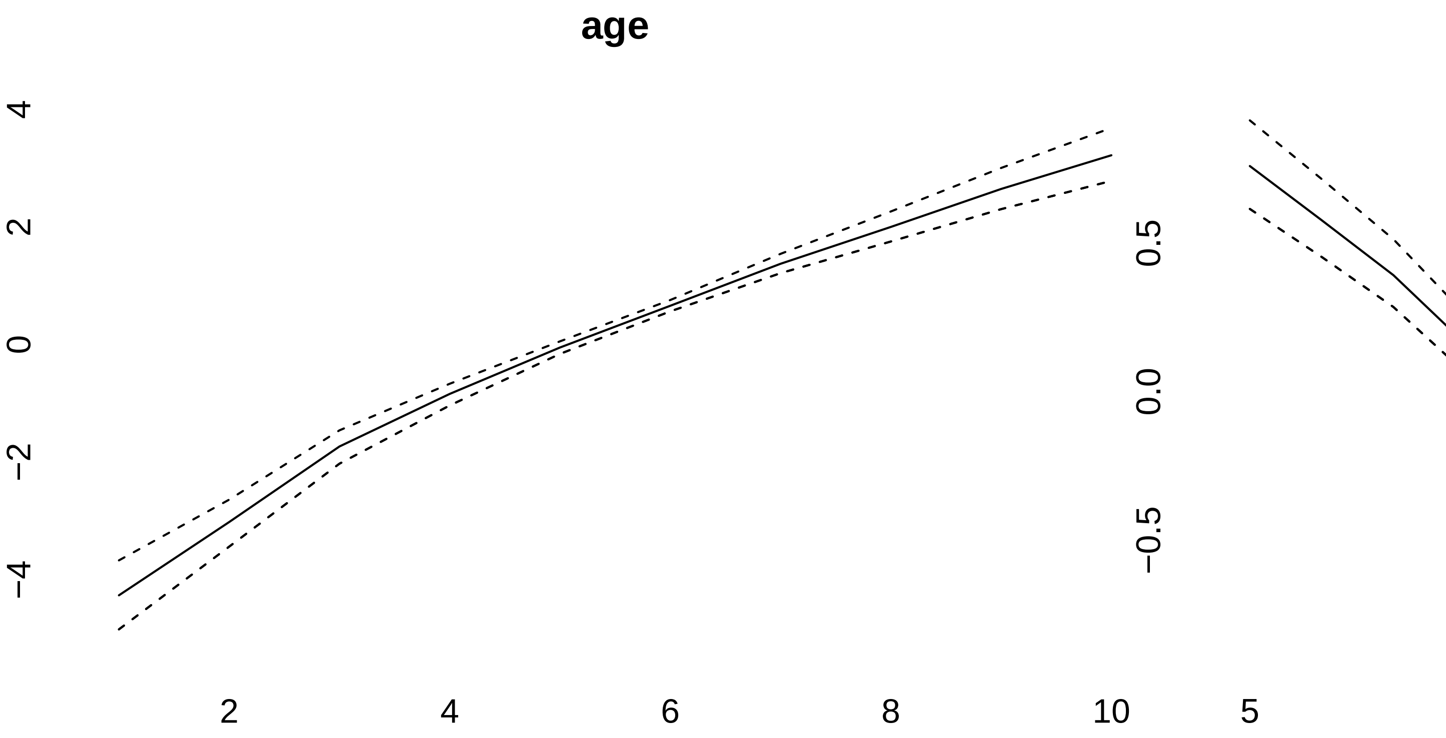
```
##
##  Model:
## age (rw1)  - period (rw1)  - cohort (rw1) model
## Deviance:     168.64
## pD:            49.10
## DIC:          217.74
##
##
##  Hyper parameters:                   5%          50%          95%
## age                               0.568        1.295        2.622
## period                           13.482       26.246       45.131
```

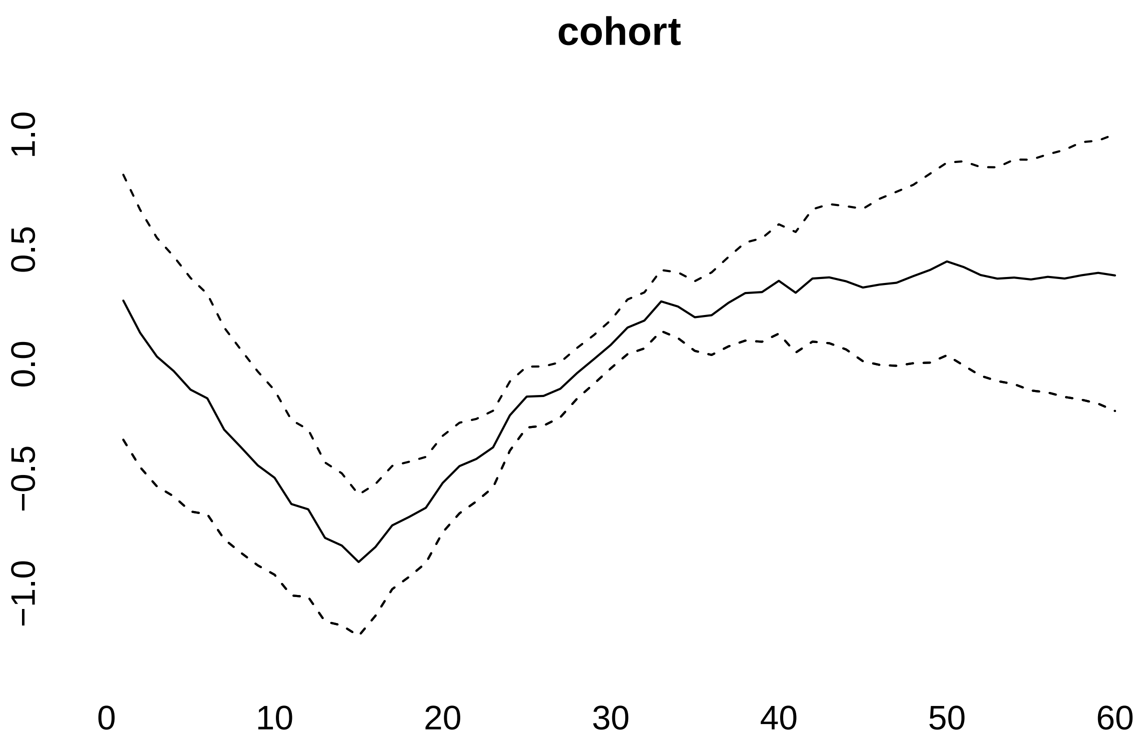
```
## cohort                              76.383      121.785      186.401
##
##
## Markov Chains convergence checked succesfully using Gelman's R (potential scale reduction factor).
```

```r
checkConvergence(simmod)
```

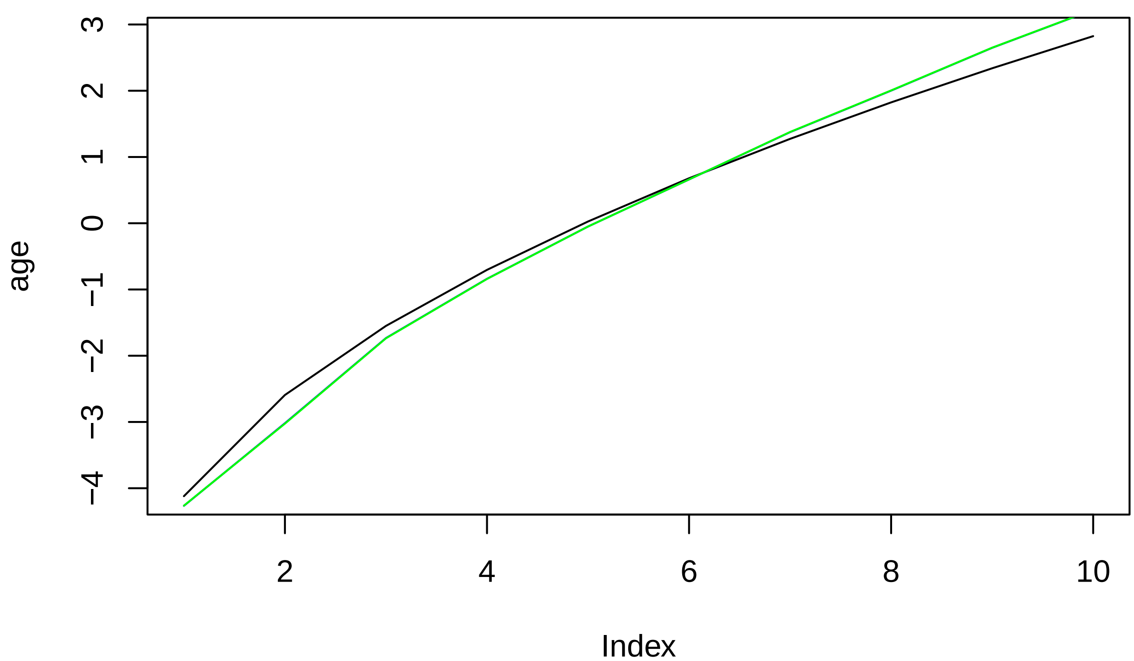
```
## [1] TRUE
```

```r
plot(simmod)
```

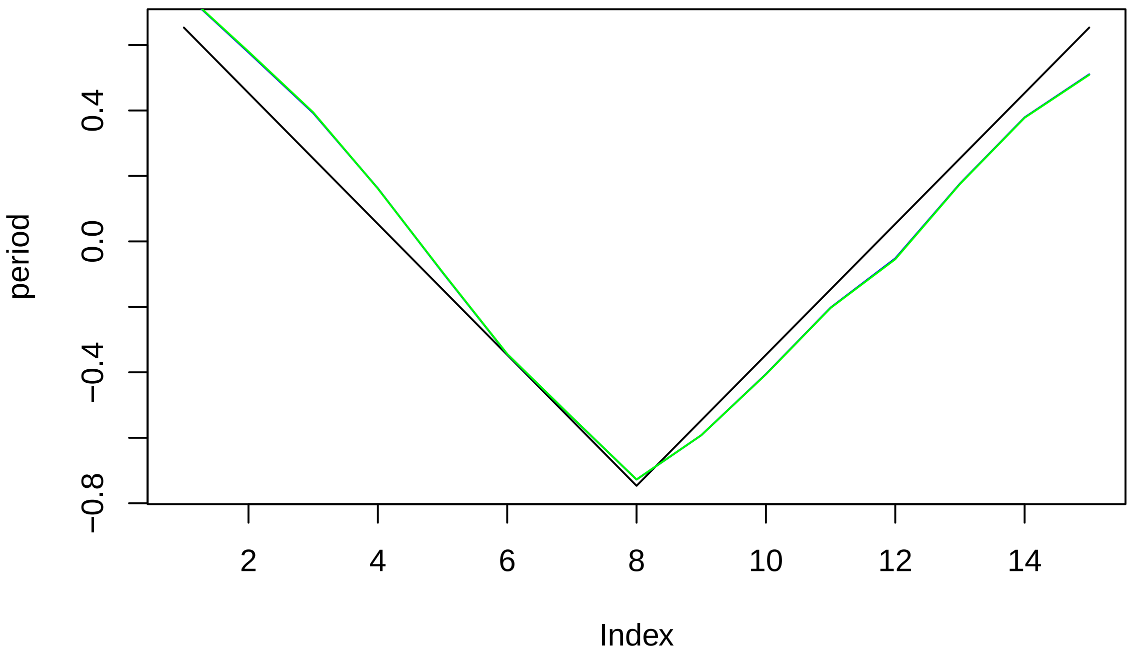




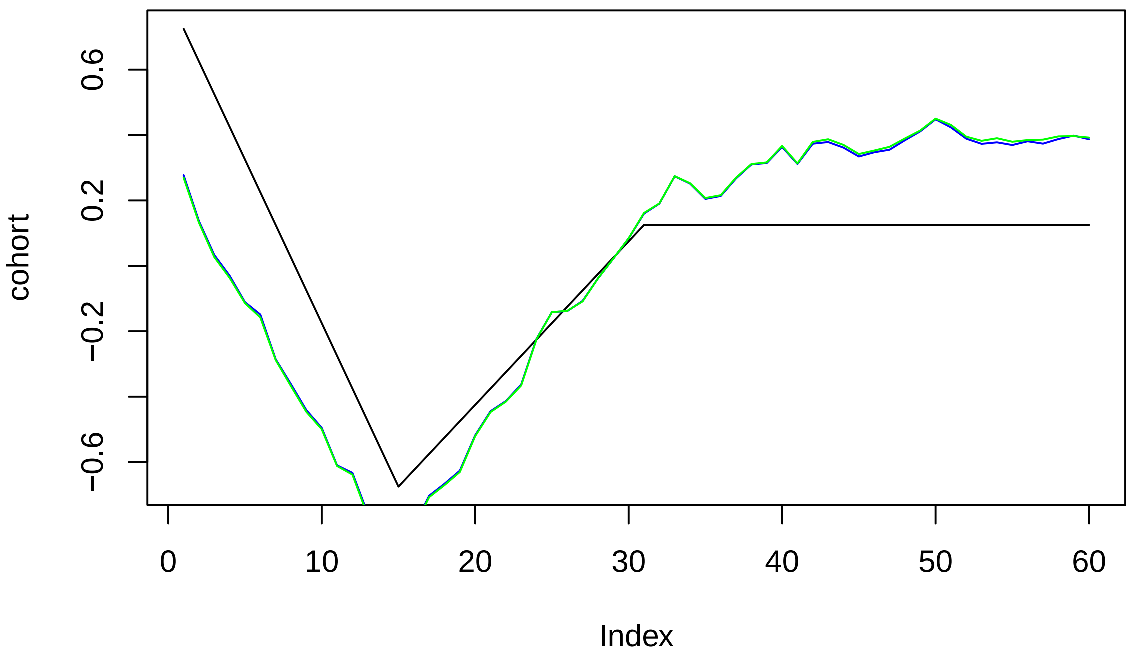

```r
effects<-effects(simmod)
effects2<-effects(simmod, mean=TRUE)
#par(mfrow=c(3,1))
plot(age, type="l")
lines(effects$age, col="blue")
lines(effects2$age, col="green")
```



```r
plot(period, type="l")
lines(effects$period, col="blue")
lines(effects2$period, col="green")
```

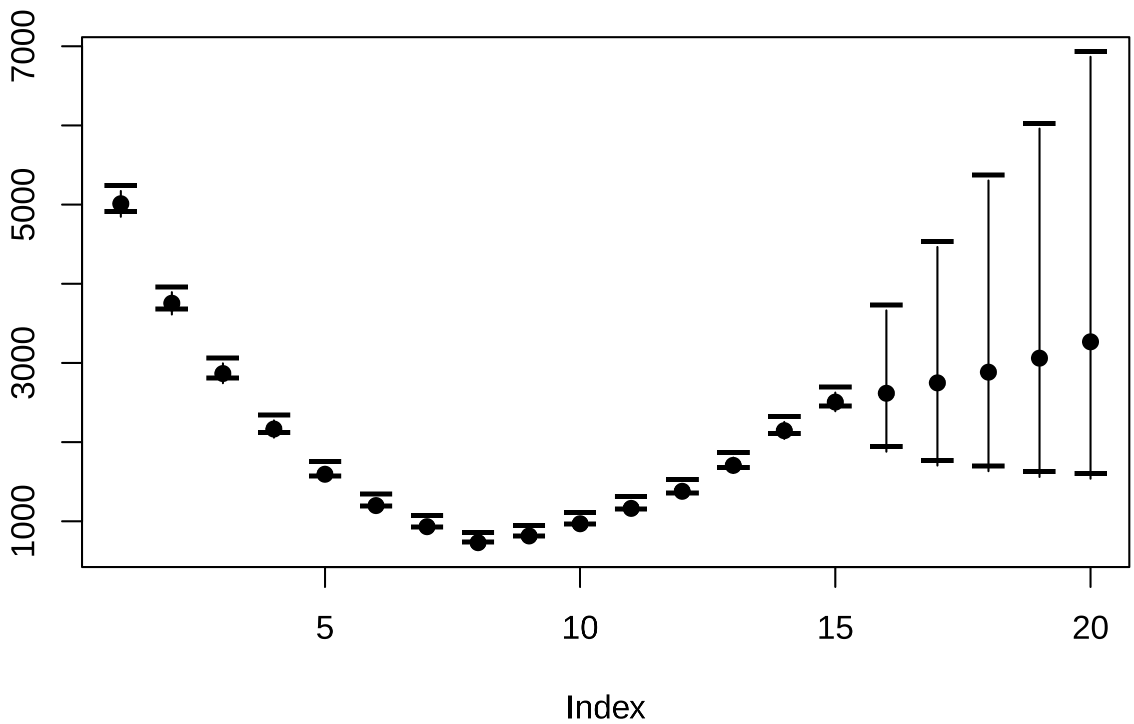



```r
plot(cohort, type="l")
lines(effects$cohort, col="blue")
lines(effects2$cohort, col="green")
```

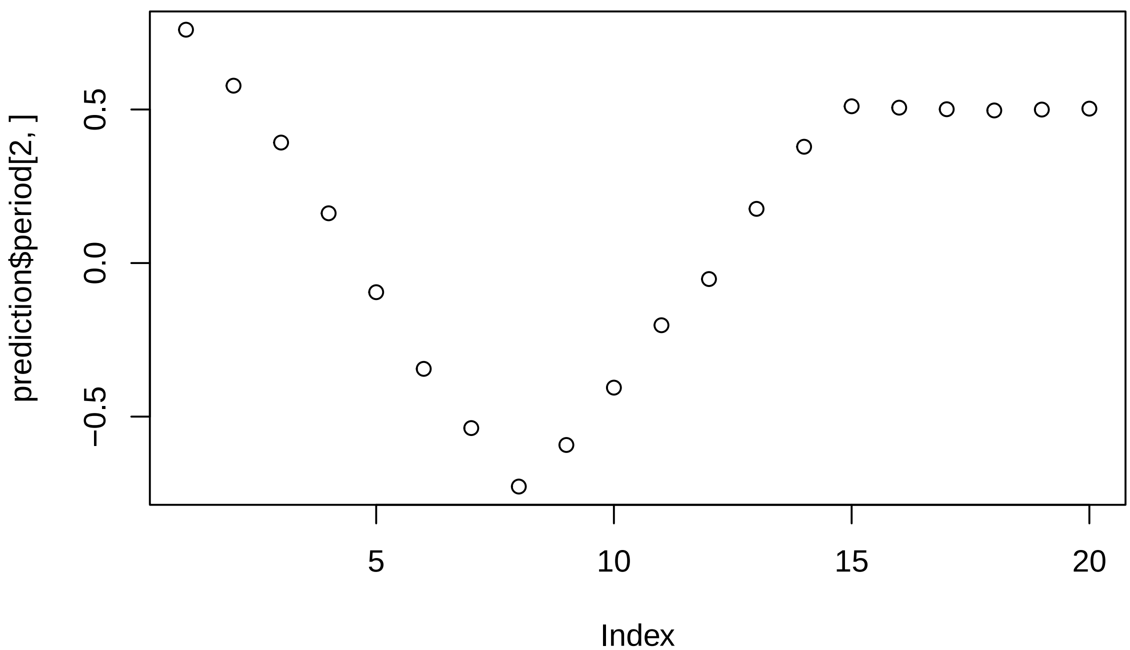



```r
prediction<-predict_apc(simmod, periods=5, population=array(1e6,c(20,10)))
```

```
plot(prediction$cases_period[2,], ylim=range(prediction$cases_period),ylab="",pch=19)
points(prediction$cases_period[1,],pch="-",cex=2)
points(prediction$cases_period[3,],pch="-",cex=2)
for (i in 1:20)lines(rep(i,3),prediction$cases_period[,i])
```




```
plot(prediction$period[2,])
```



```r
cov_p<-rnorm(15,period,.1)
```

```r
simmod2 <- bamp(cases = simdata$cases, population = simdata$population, age = "rw1",
period = "rw1", cohort = "rw1", periods_per_agegroup =periods_per_agegroup,
period_covariate = cov_p)
```

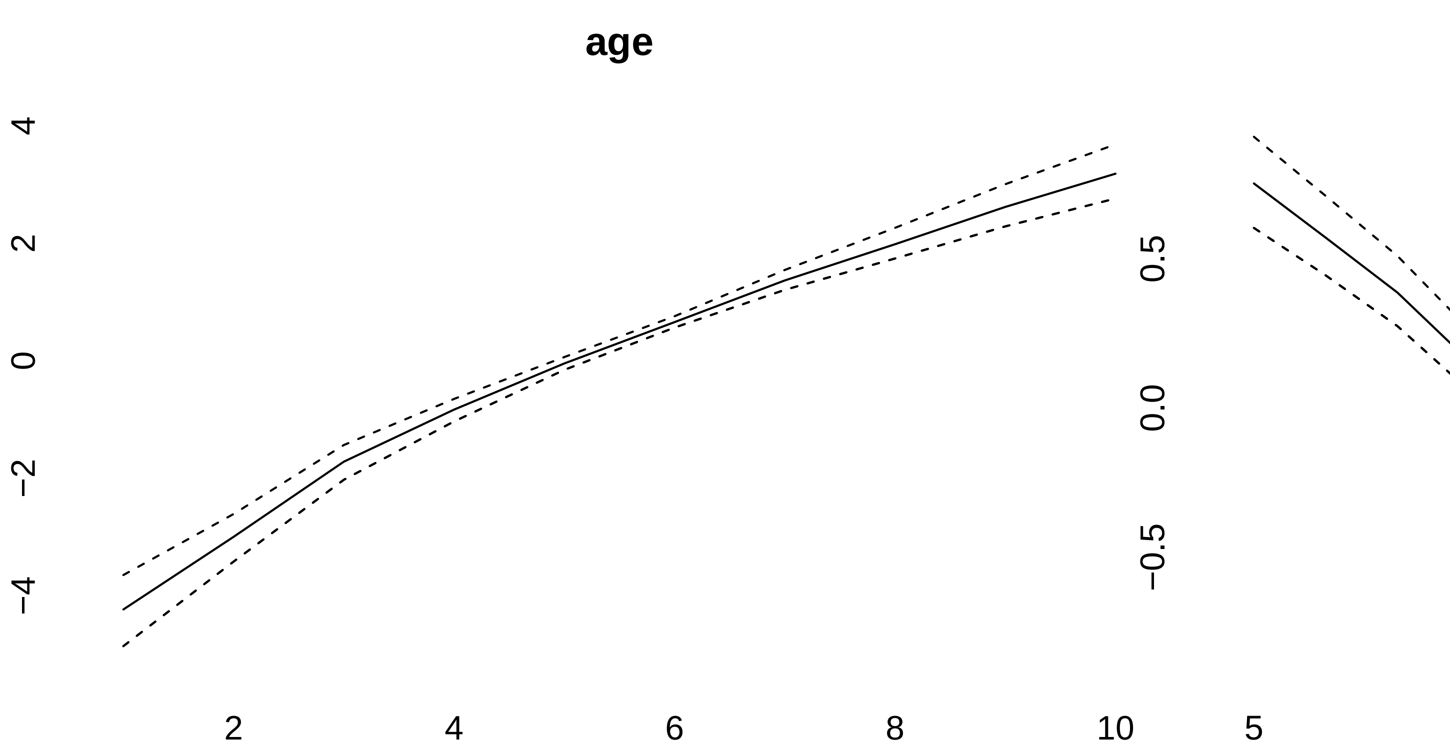
```r
print(simmod2)
```

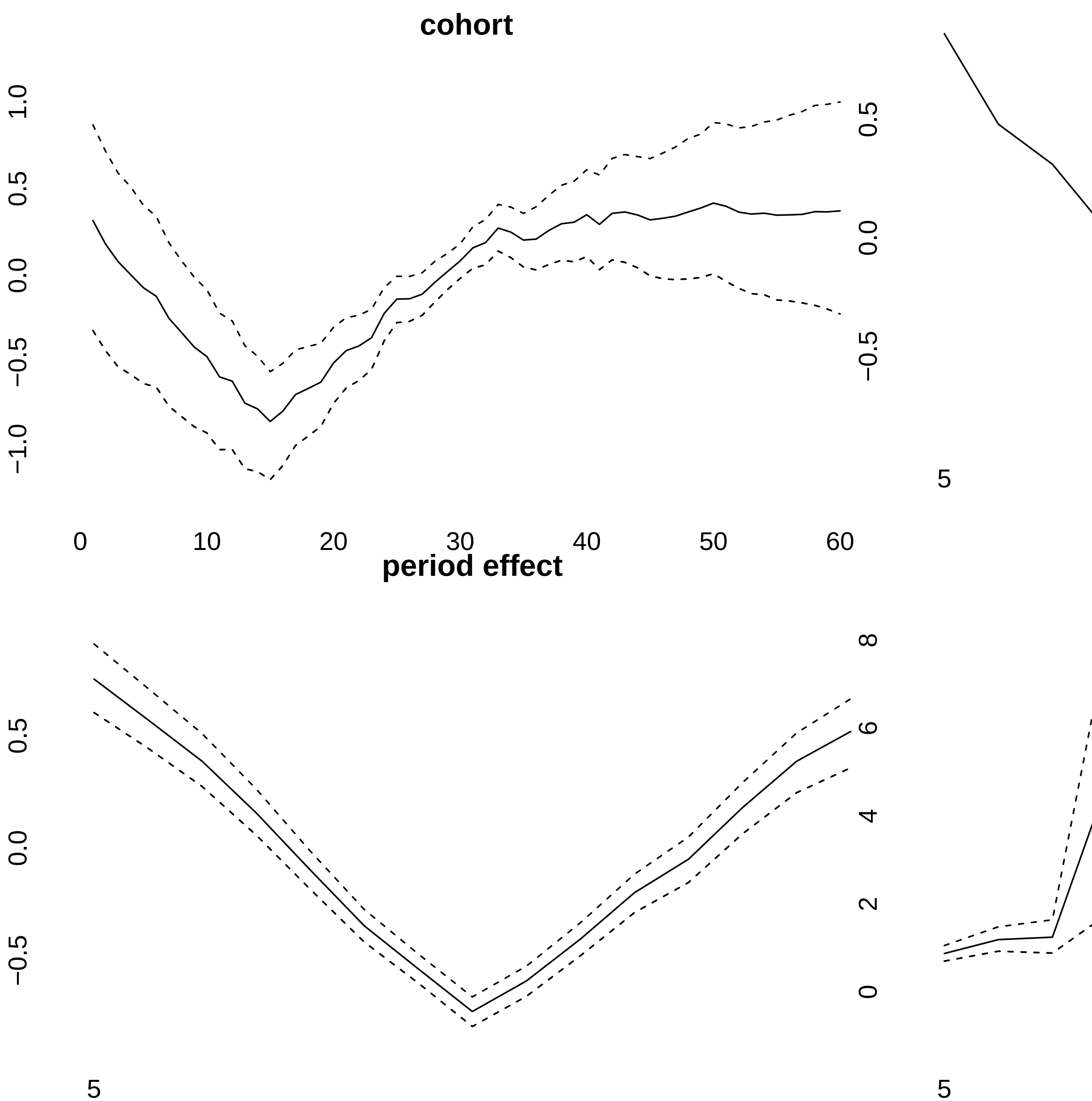
```
##
##  Model:
## age (rw1)  - period (rw1)  - cohort (rw1) model
## Deviance:     168.61
## pD:            49.05
## DIC:          217.66
##
##
##  Hyper parameters:                    5%          50%          95%
## age                                0.554        1.338        2.682
## period                            13.490       26.266       45.209
## cohort                            74.855      121.279      184.646
##
##
## Markov Chains convergence checked succesfully using Gelman's R (potential scale reduction factor).
```

```r
checkConvergence(simmod2)
```

```
## [1] TRUE
```

```r
plot(simmod2)
```




cohort
1.0
0.5
0.0
−0.5
−1.0
0
10
20
30
40
50
60
0.5
0.0
−0.5
5
period effect
0.5
0.0
−0.5
5
8
6
4
2
0
5